



Perry Nuclear Power Plant
Rod Penfield
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February 24, 2025
L-25-001

10 CFR 50.73(a)(2)(v)(D)

ATTN: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

SUBJECT:
Perry Nuclear Power Plant
Docket No. 50-440, License No. NPF-58
Licensee Event Report Submittal

On December 19, 2024, Licensee Event Report (LER) 2024-004-00, "Division 3 Diesel Generator Inoperable due to Low Output Voltage Resulting in Loss of Safety Function" (ML24354A144) was submitted by Vistra Operations Company, LLC. The enclosure is Revision 1 to that report to provide the results of the causal evaluation. The updated information is denoted by revision bars located in the right-hand margin. The report is submitted in accordance with 10 CFR 50.73. There are no regulatory commitments contained in this submittal.

If there are any questions or if additional information is required, please contact Mr. Robert Oesterle, Manager – Regulatory Compliance, at (419) 321-7462.

Sincerely,

A handwritten signature in black ink, appearing to read "Rod L. Penfield". The signature is written in a cursive, flowing style.

Rod L. Penfield

L-25-001

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Enclosure:

LER 2024-004-01

cc: NRC Project Manager
 NRC Resident Inspector
 NRC Region III Regional Administrator

Enclosure
L-25-001

Licensee Event Report 2024-004-01



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollect.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name	☒ 050	2. Docket Number	3. Page
Perry Nuclear Power Plant	☐ 052	00440	1 OF 4

4. Title:
Division 3 Diesel Generator Inoperable due to Low Output Voltage Resulting in Loss of Safety Function

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
10	29	2024	2024	- 004 -	01	02	24	2025	Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or in NRC Form 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
George Dujanovic, Staff Nuclear Engineering Specialist, Regulatory Compliance	(440) 280-5200

13. Complete One Line for Each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On October 29, 2024, at 1152, with the reactor in Mode 1 and 100 percent rated thermal power, Operations declared the Division 3 diesel generator (DG) Inoperable per Technical Specification (TS) 3.8.1, due to output voltage dropping below the TS acceptance value during a monthly surveillance run. Subsequent troubleshooting found potentiometers on the voltage regulator were oxidized, requiring exercise, which remediated the output voltage issue. Satisfactory testing of the DG, following the troubleshooting, resulted in a proper and steady output voltage, and the Division 3 DG was declared Operable on October 31, 2024, at 0030.

This is a supplemental report to LER 2024-004-00 to update the causal investigation and the corrective actions.

The cause was determined to be oxidation of the voltage regulator R1 and R3 potentiometers due to high humidity. Corrective actions include exercising potentiometer R1, R3, and R4 on the output voltage regulator to remove oxidation and revising operating procedures to perform cycling the output voltage adjustment monthly.

This event was reported on October 29, 2024, as a loss of safety function, due to the loss of the Division 3 DG, under 10 CFR 50.72(b)(3)(v)(D), for accident mitigation.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Perry Nuclear Power Plant	☒ 050	2. DOCKET NUMBER 00440	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 004	REV NO. - 01

NARRATIVE

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].

BACKGROUND:

The high pressure core spray (HPCS) [BG] system power source, the Division 3 DG [DG], is self-contained, except for access to the preferred source of offsite power through the onsite AC power distribution system and the system actuation signal source. The system is operable as an isolated system independent of electrical connection to any other system by using the Division 3 DG.

HPCS provides and maintains an adequate coolant inventory inside the reactor vessel to limit fuel cladding temperatures in the event of breaks in the reactor coolant pressure boundary. The system is initiated by either high pressure in the drywell or low water level in the reactor vessel [RCT]. It operates independently of all other systems over the entire range of pressure differences from greater than normal operating pressure to zero. The HPCS cooling decreases vessel pressure to enable the low-pressure cooling systems to function. The HPCS system pump [P] motor [MO] is powered by a diesel generator if auxiliary power is not available; the system may also be used as a backup for the reactor core isolation cooling (RCIC) [BN] system.

DESCRIPTION OF EVENT:

On October 29, 2024, at 1152, with the reactor in Mode 1 and 100 percent rated thermal power, during a preventative maintenance run of the Division 3 DG, it was noticed that the generator output voltage had dropped below the Technical Specification (TS) acceptance value of 3994 volts (3900 volts, adjusted for instrument inaccuracy). The lowest value recorded during the DG run was 3947 volts. The Division 3 DG was declared Inoperable, but available, and TS 3.8.1, AC Sources – Operating, Condition B, one required DG inoperable, was entered.

During troubleshooting, it was discovered that an oxidation layer had built up on the R1 potentiometer in the output voltage regulator [RG]. Exercising the potentiometer throughout its range caused the oxidation layer to be removed, thereby eliminating the high resistance point and allowing the voltage regulator to function normally. A post-maintenance run of the Division 3 DG was performed resulting in a normal and steady output voltage. On October 31, 2024, at 0030, the Division 3 DG was declared Operable and TS 3.8.1 was met.

Due to the Inoperability of the Division 3 DG, it was determined to be an event or condition that could have prevented fulfillment of a safety function under 10 CFR 50.72(b)(3)(v)(D) for mitigating the consequences of an accident during a loss of offsite power/loss of coolant accident (LOOP/LOCA).

The NRC was notified at 1732, on October 29, 2024, and this was documented on Event Notification 57406.

No other failures of the Division 3 DG or the voltage regulator could be found during the previous three-year timeframe. Although outside of the three-year timeframe, a previous similar event was identified and is discussed later in this report.



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	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 004	REV NO. - 01

NARRATIVE

CAUSE OF EVENT:

The cause for the voltage fluctuations was oxidation on the R1 and R3 potentiometers in the voltage regulator.

The R1 and R3 potentiometers were oxidized due to high humidity caused by water intrusion into the diesel room in June of 2024 from a penetration leak in the ceiling.

A contributing cause to the event was determined to be that the maintenance strategy to clean/inspect the R1, R3, and R4 potentiometers was performed every 4 years. EPRI Technical Report 1011109 advises checking the condition of these potentiometers every 2 years.

CORRECTIVE ACTIONS:

Completed Actions:

- Division 3 DG voltage regulator potentiometers R1, R3, and R4 were exercised, and the DG was started and ran with no voltage fluctuations noted.
- Curbs have been installed as barriers around the DG roof penetrations to prevent rainwater from entering the DG areas.

Scheduled Action:

- Revise the diesel operating procedure and the testing surveillances to cycle the voltage regulator adjustment in the control room after running the engine with no load prior to shutting the engine down to help prevent oxidation of the R1 potentiometer.
- The maintenance task to clean/inspect the R1, R3, and R4 potentiometers will be changed to be performed every 2 years, to align with the EPRI report.

EVENT ANALYSIS:

A probabilistic risk assessment (PRA) evaluation was performed for the October 29, 2024, Division 3 DG voltage instability event. The DG remained available, due to the momentary voltage drop below the TS acceptance value of 3994 volts (3900 volts, adjusted for instrument inaccuracy), and therefore there is no risk impact to the PRA. The risk of this event is therefore considered to be of very low safety significance in accordance with the Regulatory Guidance. Site auxiliary power was available during the Inoperability period and RCIC remained Operable.

An engineering evaluation concluded that the voltage regulator providing a bus voltage of 3947 volts on 10/29/24 would not have prevented the high pressure core spray system from accomplishing its safety function based upon the existing design bases analysis requiring a minimum steady-state bus voltage of 3730 volts to the connected loads. Therefore, in accordance with NEI 99-02, Regulatory Assessment Performance Indicator Guideline, Section 2.2, Mitigating Systems Cornerstone for Safety System Functional Failures, this event will not be counted against the Reactor Oversight Process (ROP) Safety System Functional Failure Performance Indicator.



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NARRATIVE

PREVIOUS SIMILAR EVENT:

A previous similar event occurred on June 1, 2021. Division 3 DG output voltage fluctuated below the technical specification limit during a preventative maintenance run of the Division 3 DG. The voltage regulator was replaced. Laboratory testing found an instability in the linearity of the resistance of potentiometer R3 when turned. Research found that this is a known failure mode with the SR8A line of voltage regulators causing voltage instabilities.